

Lesson 3 — Decomposition — Break It Down

Matches LEGO Teacher Portal lesson: "Choreograph with Color".

Objective (I can...): I can break a big task into smaller steps before I code it. **CS concept:** Decomposition (splitting a problem into smaller parts) · **Builds on:** Lesson 2 (sequences).

AI Connection (2 min): To "recognize a cat," AI breaks the picture into tiny parts — edges, ears, whiskers — then combines them. *First-principles question:* "What small clues would you check to know it's a cat and not a dog?"

Materials: LEGO CS & AI kit + device per pair, sticky notes or mini whiteboard per pair. **Model reuse:** use the Basics model built in Lesson 2 (no rebuild) — today is code plus one tiny build tweak.

60-minute flow - Warm-up (8): "Program the human robot" — one kid is the robot and stands across the room; the partner gives only literal steps ("step forward, step forward, turn right") to guide them across. If a step is vague, the robot freezes. Point: machines need small, exact steps — that's decomposition. - **Concept + demo (10):** Coach picks a behavior (make it wave hello), says the small steps aloud, then turns each into a block. (*Rotate the format: try "mystery program — guess the steps before I run it."*) - **Build/Code (30):** Using the already-built model, pairs write small steps on sticky notes first, then place one block per step and run. **Make-it-yours beat:** add one small model change — an attachment or a LEGO "face" — and name the bot. *Swap at 15 min.* - **Challenge (7):** Add one more small step (light or sound) and note where it fits in the list. - **Share + pack-up (5):** Pairs show their sticky-note list next to the running (and now named) program. Model stays built.

Pair roles: Navigator writes the sticky-note steps; Driver codes them. *Swap at 15 min. (Trio: third child reads the step list back during testing.)*

Differentiation: *Support* — a half-filled step list to complete. *Extension / fast-finisher* — decompose a two-behavior task. *Reteach* — act out the behavior physically, naming each step, then map to blocks.

Exit check: Pair's written step list matches the blocks in their working program. **At-home (optional):** Break "make a bowl of cereal" into 6+ small steps. ~10 min.